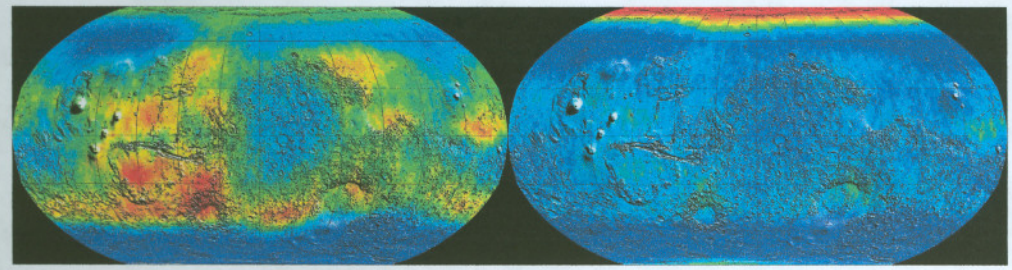
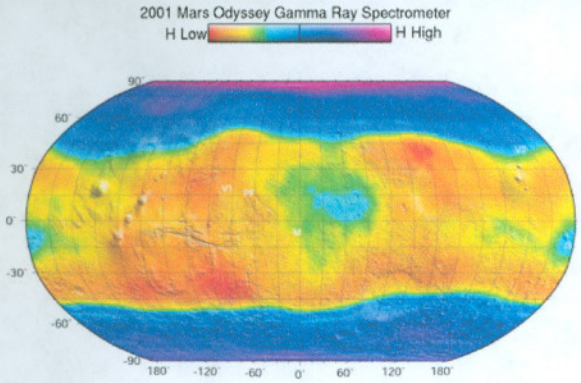
Analytical Cumulative Examination
November 10, 2006





Shown above are maps of thermal neutrons (top left), epithermal neutrons (top right) and gamma radiation (bottom) detected by the Mars Odyssey spacecraft.

1. Discuss the instruments that made these measurements. Be as precise as you can about the scientific principles of their operation and discuss any practical details of their operation.
2. Discuss the mechanisms by which these radiations are produced by Mars. Again, credit will be given for details – especially if expressed in a quantitative fashion – rather than vague generalizations. Diagrams and tables will help in this discussion.
3. Discuss the interpretation of the data obtained. What can you learn from the plots reproduced above about past and present Mars.
4. If you were given the ability to land at three locations on Mars – based on the information – where would you land and what measurements would you make. Keep in mind that the mission and the measurements must be driven by science issues.
5. If you were given the ability to land and recover samples for return to earth where would you land and what measurements would you make on Earth. Keep in mind that the mission and the measurements must be driven by science issues.

Each question will be weighted equally.